AI Resume Ranker

Smart Resume Screening using Natural Language Processing

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Problem Statement

Manual Screening Bottleneck

Companies receive hundreds of applications for each position, making manual review impractical and time-intensive.

Time-Consuming Process

HR teams spend excessive hours reading through resumes instead of conducting actual interviews.

Need for Automation

Recruitment departments need intelligent systems to automate initial screening and prioritize qualified candidates.

Project Objective

Build an AI-Powered Solution

Develop an intelligent system that automatically ranks resumes based on their alignment with job descriptions using advanced Natural Language Processing techniques.

Key Goals

- Improve hiring efficiency by 70%+
- Increase candidate selection accuracy
- Reduce manual screening time
- Provide data-driven insights

Technology Stack

Frontend

- HTML5 for structure
- CSS3 for styling
- Responsive design

Backend

- Python programming
- Flask framework
- RESTful API

AI Libraries

- spaCy for NLP
- Pandas for data
- Scikit-learn for ML

Processing

- PyPDF2 for extraction
- Text processing
- Data cleaning

How It Works

01

User Inputs Job Description

Hiring manager enters detailed requirements for the position

02

Upload Multiple Resumes

System accepts multiple PDF resumes simultaneously

03

Extract Text from PDFs

PyPDF2 converts document format to processable text

04

NLP Processing & Comparison

spaCy analyzes and compares content with job requirements

05

Generate Similarity Scores

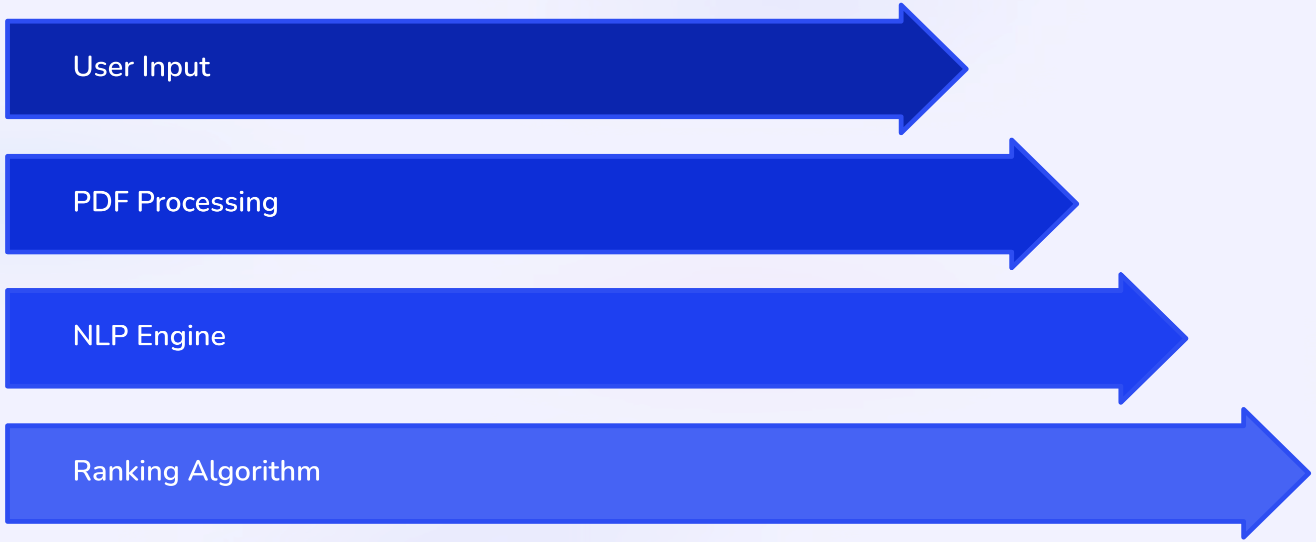
Algorithm calculates match percentages for each candidate

06

Display Ranked Results

Sorted list shows best-matching candidates at top

System Architecture



Key Components

The system follows a streamlined pipeline from data ingestion through intelligent processing to actionable output.

- **Input Layer:** User interface for data entry
- **Processing Engine:** PDF extraction and text cleaning
- **NLP Core:** Semantic analysis and matching
- **Ranking Module:** Score calculation
- **Output Interface:** Results presentation

Key Features

Multiple Resume Upload

Batch processing allows simultaneous upload of dozens of resumes in PDF format for comprehensive evaluation.

NLP-Based Matching

Advanced natural language processing understands context, skills, and experience rather than just keyword matching.

Intelligent Ranking

Dynamic scoring system ranks candidates by match percentage, prioritizing most qualified applicants automatically.

CSV Export

Download complete results with scores for further analysis in spreadsheets or HR systems.

User-Friendly Interface

Clean, intuitive design makes the system accessible to non-technical HR professionals.

Fast Processing

Processes multiple resumes in seconds, dramatically reducing screening time compared to manual review.

Output & Results

Ranked Results Table

The system generates a comprehensive comparison showing each candidate's match score, skills alignment, and ranking position.

Score-Based Comparison

Each resume receives a percentage score indicating match quality, allowing easy comparison between candidates.

Challenges & Future Enhancements

Challenges Faced

PDF Text Extraction

Handling diverse PDF formats and layouts required robust error handling

NLP Accuracy

Tuning spaCy models for consistent semantic understanding

Model Setup

Initial spaCy installation and configuration presented technical hurdles

Future Enhancements

ATS Score Visualization

Interactive charts showing score breakdown by categories

AI Skill Extraction

Automated identification of technical skills and certifications

Analytics Dashboard

Comprehensive metrics on hiring pipeline performance

Cloud Deployment

Scalable infrastructure for enterprise-level usage

Thank You

Project Summary

This AI-powered system automates resume screening, saving time and improving hiring decisions through intelligent NLP-based matching.

Key Achievements

- Built complete NLP pipeline
- Integrated multiple libraries
- Created intuitive interface
- Achieved accurate matching

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